Seagrass sediment organic carbon burial rates are globally significant

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Abstract

Seagrass ecosystems are pivotal contributors to coastal carbon sequestration through the long-term burial of organic carbon (OC) in sediments. Yet global burial estimates remain uncertain, with early values of 138 ± 38 g OC m-2 yr-1 derived from a limited dataset biased toward highly depositional communities and indirect, production-based approaches. We call for a downward revision, supported by a data-driven assessment based on a global synthesis of 326 dated sediment cores integrating OC burial over the last century. We find that seagrass meadows bury OC at a geometric mean rate of 26 ± 2 g OC m-2 yr-1, with an area-weighted global average of 33 ± 10 g OC m-2 yr-1 accounting for bioregional differences in seagrass distribution. Globally, these rates scale to 6–16 Tg C yr-1, based on current mapped seagrass extent (247,800–366,200 km2), and reveal that ~15% of seagrass net community production is retained and buried locally. Although our estimate is roughly one-fourth of earlier values, seagrass sediments still account for 3–6% of total oceanic OC burial, despite occupying <0.1% of the seafloor. Combined with mangroves and tidal marshes, OC burial in vegetated coastal sediments represents an estimated 8–13% of total oceanic OC burial.

Main text

The documented global extent of seagrass ecosystems is less than 0.1% of the area of the coastal ocean, yet they are estimated to contribute 10-18% of its total organic carbon (OC) burial[1]. Earlier assessments, mainly indirect, estimated that seagrasses bury OC at an average rate of 138 ± 38 g OC m^{-2} yr^{-1} [2], comparable to other blue carbon ecosystems[3,4] and roughly 30 times higher than rates in terrestrial forest soils[2]. Their global extent and their capacity to capture and bury OC over the long-term have positioned seagrasses as a supportive element of nature-based climate solutions, with protection and large-scale restoration estimated to avoid up to ~1% of annual global greenhouse gas emissions[5]. This potential could be larger in countries that have extensive seagrass coverage and relatively low anthropogenic emissions[6]. Although interest in including seagrass management in Nationally Determined Contributions (NDCs) is growing, only 21 countries explicitly reference seagrasses[7], and just a few (e.g., Australia [8], Japan [9])report them in their United Nations Framework Convention on Climate Change (UNFCCC) Greenhouse Gas inventory (GHGI), despite IPCC guidelines enabling such reporting[10].

Climate change mitigation opportunities based on seagrass management require better understanding of their global distribution, loss and recovery trajectories, and their rate of OC burial in sediments[5,11]. Past attempts at directly estimating global OC burial in seagrass sediments relied on six radiocarbon-dated rates from large *Posidonia oceanica* mattes[12,13], and a range of indirect approaches, including mass balance[14], biomass accretion[15], or sediment traps[16]. These limited measurements were further supplemented by available estimates of seagrass net community production (NCP)[17] and burial of allochthonous OC[18] to finally result in the hitherto accepted value of 138 ± 38 g OC m^{-2} yr^{-1} [2].

The sparse abundance of direct estimates of OC burial rates in seagrass sediments contrasts with the greater number of estimates on sediment OC stock size available (> 2700)[19]. Depth-integrated sedimentary OC stocks, paired with models or assumptions of OC lability, can inform about the amount of OC that could be remineralized and released to the atmosphere if seagrass meadows were degraded[20,21]. However, they do not capture the carbon burial efficiency of seagrass ecosystems, or their climate mitigation potential over the coming decades (50–100 years). The time required for seagrass sediments to accumulate a given OC stock can vary by more than an order of magnitude[22,23], challenging the commonly held assumption that large OC stocks indicate high OC burial efficiency[24]. Indeed, criticism has arisen over the practice of using OC stocks to infer sequestration[25,26], highlighting the need to consolidate direct estimates of OC burial rates in seagrass sediments.

Global assessments of OC burial rates for other coastal blue carbon ecosystems have been revisited and refined in recent years [3,4]. In tidal marsh and mangrove ecosystems, global OC burial rates have been reassessed through synthesis of studies that combined sediment OC content with sedimentation rates determined using radiometric dating, primarily ^{210}Pb and ^{137}Cs, and to a lesser extent, using marker and event horizons of known dates [4,27,28]. Beyond enabling the calculation of OC burial rates, these methods have allowed the identification of settings with no net sediment accumulation[29], reinforcing the need to assess burial rates, rather than rely solely on stock data. Radiometric dating of seagrass sediments to derive OC burial rates has several challenges [30]. Primarily, sediment mixing, which is common in seagrass sediments, may overestimate OC burial rates or, in some cases, preclude their determination altogether. Secondly, the predominantly mineral and sandy nature of seagrass sediments [31] can dilute radionuclide concentrations, limiting the depth and timescale over which reliable age models can be developed, potentially leading to a substantial bias in published OC burial rates toward muddier depositional environments.

The rise of blue carbon strategies has greatly stimulated research into OC burial by seagrasses, resulting in numerous site-specific studies across seagrass species and geographic regions. These newer studies have generally reported lower burial rates than previously estimated for seagrasses globally [32]. Nevertheless, a comprehensive and updated synthesis of seagrass OC burial remains lacking, and available estimates are still scattered throughout the literature. Given the importance of seagrass meadows to the oceanic carbon budget [1], we synthesized contemporary OC burial rates on the century scale in seagrass sediments to provide an updated global estimate and assess global patterns. Our analysis integrated downcore sediment profiles of OC content and ^{210}Pb specific activities along with ancillary variables and relevant metadata, resulting in a comprehensive, open-source dataset. These data were integrated with global classification schemes such as seagrass bioregions[33] and nearshore coastal geomorphological typologies[34], alongside seagrass genus, to understand its variability. Consolidating knowledge on seagrass OC burial rates contributes to refining the global ocean carbon budget, validates estimates of seagrass OC fluxes, and supports the revision of the global default (Tier 1) value for use in national GHGI reported to the UNFCCC.

The dataset comprised 326 sediment profiles from seagrass ecosystems across a latitudinal range of 64ºN to 35ºS (Fig. 1), including vegetated (89%) and unvegetated (11%) sediments. Overall, a total of 222 records from our primary dataset reported quantifiable OC burial rates (210 from vegetated and 12 from unvegetated sediments). In the remaining 104 cores, OC burial rates could not be resolved equally due to intense sediment mixing extending well below the surface and the absence of net sediment accumulation. The latter was inferred from the absence of excess ^{210}Pb, the unsupported fraction of ^{210}Pb derived from atmospheric fallout, which accumulates on the seafloor over time and serves as a tracer for recent sediment deposition[35].

Consistent with reported OC stock assessments[19,20,36], data were geographically concentrated in Australia, North America, and the Mediterranean Sea, while data were sparse for South America, Africa, South and Southeast Asia, despite the extensive seagrass distribution in the latter. By bioregion, 10% of the estimates originated from the North Atlantic, while 21 and 26% came from the Tropical Atlantic and Southern Oceans (primarily from the tropical Western Atlantic, and Australia, respectively), with the remaining regions contributing ~15% each.

Organic carbon burial rates in seagrass sediments

Global variability in OC burial rates within vegetated sediments with net positive accumulation ranged over 3 orders of magnitude, from 1.3 ± 0.5 g OC m^{-2} yr^{-1} (Ameralik, Greenland [38]) to 976 ± 132 g OC m^{-2} yr^{-1} (Swan Lagoon, Shandong, China[22]). Regional and local ranges were similarly pronounced within bioregions, with over two orders of magnitude variability within the N. Atlantic and N. Pacific, and an order of magnitude in the Mediterranean, the Tropical-Indo Pacific, Tropical Atlantic and the Australian portion of the Southern Oceans (Table S1).

The distribution of global OC burial rates in seagrass sediments with net accumulation was log-normal with a strong skew towards low values (Fig. 2b). Under these conditions, the geometric mean provided the appropriate measure of central tendency and was therefore used (Table 1). Across the dataset, global contemporary OC burial rates in seagrass sediments averaged 26 ± 3 g OC m^{-2} yr^{-1} (95% CI: 22–29; n = 203), a value that remained unchanged even after excluding records affected by intense sediment mixing (27 ± 2 g OC m^{-2} yr^{-1}; 95% CI: 23–31; n = 123), known to overestimate rates[30]. Including sites with no net sediment accumulation, suggesting negligible burial, significantly lowered the estimate to 19 g OC m^{-2} yr^{-1} (95% CI: 16–23; n = 250) (W = 30146, $p < 0.001$).

Organic C burial rates measured in seagrass sediments from the Tropical Atlantic were the highest (Kruskal–Wallis test, H = 28.5, df = 5, $p < 0.05$), while no significant differences were observed in OC burial rates among the other five bioregions ($p > 0.05$) (Fig. 2a). Considering the differences in seagrass distribution across bioregions, we estimated an area-weighted average OC burial rate of 33 ± 10 g OC m^{-2} yr^{-1} (95% CI: 25–44) in seagrass sediments with net accumulation, which was not significantly different from the geometric mean OC burial rate calculated across all vegetated sediments showing net positive accumulation (Fig. 2b).

Table 1. Summary of organic carbon (OC) burial rates, sediment mass accumulation rates (MAR), and OC content in seagrass sediments dated to the last century. The 'use value' represents measures of central tendency with 95% confidence intervals. The geometric mean was used for variables with log-normal distributions, while the median was used for variables that did not follow a normal distribution, even after log transformation. Summary statistics by bioregion can be found in Table S1.

Parameter		n	Range	Mean	SD	Median	Geom. Mean	Use Value
OC Burial rate ($g\ C\ m^{-2}\ yr^{-1}$)	Global	202	1.3-976	44	80	25	26	Geom. Mean: 26 95% C.I.: 22 - 29
	w/ negligible accumulation	249	0-976	36	74	19	-	Median: 19 95% C.I.: 16 - 23
MAR ($g\ cm^{-2}\ yr^{-1}$)	Global	202	0.02-2.8	0.33	0.36	0.22	0.22	Geom. Mean: 0.22 95% C.I.: 0.19-0.24
	w/ negligible accumulation	249	0-2.8	0.27	0.35	0.17	-	Median: 0.17 95% C.I.: 0.14–0.19
OC (%)	Global	203	0.09-9.5	1.7	1.6	1.2	1.1	Geom. Mean: 1.1 95% C.I.: 1.0–1.3
	w/ negligible accumulation	250	0.05-9.5	1.5	1.5	0.91	0.94	Geom. Mean: 0.94 95% C.I.: 0.8–1.1

Compared to the arithmetic mean global OC burial rate reported by McLeod et al.[2] (138 ± 38 g OC $m^{-2}\ yr^{-1}$), our updated arithmetic mean for vegetated sites with net positive accumulation (44 ± 80 g OC $m^{-2}\ yr^{-1}$) was 68% lower. In terms of geometric means, our estimate (26 ± 3 g OC $m^{-2}\ yr^{-1}$) was ~40% lower than the IPCC Tier 1 global default value for seagrass OC burial (geometric mean 43 g OC $m^{-2}\ yr^{-1}$, 95% CI: 20–70)[10], while our area-weighted geometric mean (33 ± 10 g OC $m^{-2}\ yr^{-1}$) was ~20% lower, though both estimates fell within the IPCC 95% confidence interval. When sites with negligible accumulation were included, the global average OC burial rate was ~75% lower than the global arithmetic mean from McLeod et al.[2] , while the median global rate was ~50% lower than the IPCC geometric mean (Fig. 3a).

This downward revision resulted from the inclusion of 50 times more direct estimates of sediment OC burial than previous global syntheses, all standardized to a common timescale and benefiting from broader data coverage. Unlike earlier global assessments, our direct analysis captures the mean *in situ* accumulation of OC in seagrass sediments over the last century, excluding export and burial beyond the meadows, sequestration in biomass or net ecosystem uptake. The expanded and standardized dataset highlights that early estimates were biased toward habitats with high OC burial rates and inflated by the mixing of methodological approaches. Although the habitat bias was also evident in early ^{210}Pb-derived rates, the growing number of measurements since 2018 has led to stable mean and median estimates and reduced standard errors (Fig. 3a), suggesting that the current number of measurements is sufficient to provide a robust, revised, global average of OC burial in seagrass sediments for blue carbon assessments and IPCC Tier 1 estimates.

On a per-area basis, the geometric mean and median OC burial in seagrass sediments were about one-fifth those of tidal marshes (geom. mean: 124 ± 9 g OC m^{-2} yr^{-1}; median: 133 g OC m^{-2} yr^{-1})[3] and mangroves (geom. mean: 121 ± 19 g OC m^{-2} yr^{-1}; median: 136 g OC m^{-2} yr^{-1})[4]. This challenges previous assertions that seagrasses bury, on average, OC at rates comparable to mangroves and tidal marshes, yet shows that seagrass sediment OC burial per unit area is still approximately six to eight times higher than in terrestrial forest soils (Fig. 3b).

Interplay between organic carbon content and sedimentation rates

The nearly three orders of magnitude variation in OC burial rates across seagrass cores arises from the combined variability of its two primary components, OC content and sediment mass accumulation rates (MAR), each spanning roughly two orders of magnitude (Table 1). In principle, this could produce a ~4-fold range in burial rates, but such extremes were not observed because high MAR and high OC content did not co-occur in the same sediments (Fig. 4).

Organic C content in seagrass sediments accumulated over the past century ranged from 0.09% to over 9%, with a geometric mean of 1.1% of dry weight and a global mean of 1.7% (Table 1), roughly 1.5 times higher than the median (0.7%) and mean (1.3%) values reported by Krause et al.[19] for seagrass sediments globally. The difference between our OC content estimates and those reported globally may be partially explained by the inclusion of surface samples to 1 m depth in earlier studies, without standardization to a specific depth or temporal horizon. Including seagrass records with negligible sediment accumulation over the past century in our estimate reduced the difference between our mean and median values and those reported globally [19] (Table 1).

Sediment MAR in vegetated records ranged from 0.02 to 2.8 g cm^{-2} yr^{-1} with a geometric mean of 0.22 g cm^{-2} yr^{-1} (95% C.I.: 0.19 - 0.24, n =203) (Table 1). This corresponded to a vertical accretion rate of 2.4

mm yr^{-1} (2.1–2.6, 95% C.I.), calculated using a median dry bulk density of 0.92 g cm^{-3} estimated from seagrass sediments accumulated over the last century across the dataset. However, about 16% of the records were collected from vegetated but non-depositional sites as indicated by negligible specific activities of excess ^{210}Pb. Including these non-depositional, yet vegetated records in the global MAR calculation resulted in a lower median estimate of 0.17 g cm^{-2} yr^{-1} (C.I.: 0.14–0.19), corresponding to sediment accretion rates of 1.5–2.1 mm yr^{-1}.

Sediment OC content and MAR followed a negative exponential relationship, with higher sedimentation rates diluting OC concentrations due to greater mineral input (Fig. 4). The highest sediment OC burial rates (> 43 g OC m^{-2} yr^{-1}; interquartile range, IQR: 56-119) generally occurred when MAR exceeded 0.18 g cm^{-2} yr^{-1} and OC content was > 1.6% (Table S2), though peak rates could also occur at very low MAR (~0.05 g cm^{-2} yr^{-1}) if OC content was maximal (9.5%). In contrast, the lowest OC burial rates (< 13 g OC m^{-2} yr^{-1}; IQR: 6–11) were observed at sites where either both OC content and MAR were below-average (< 0.8% and < 0.26 g cm^{-2} yr^{-1}), or where MAR was relatively high (~0.6 g cm^{-2} yr^{-1}) but OC content was extremely low (~0.09%). Intermediate OC burial rates (13–43 g OC m^{-2} yr^{-1} IQR: 19–34), observed in 50% of the sites, were associated with moderate OC content (0.7–1.9%) and MAR values between 0.11 and 0.36 g cm^{-2} yr^{-1}. Notably, at MAR below 0.1 g cm^{-2} yr^{-1}, OC content spanned the full range observed across the dataset (0.09–9.5%), highlighting the potential for slow accumulating environments to exhibit variable or high OC content and stocks.

Differences in the rate at which OC content declines with increasing MAR represent distinct seagrass depositional environments, each with its own balance of sedimentation and OC dynamics promoting OC burial (slopes in Fig. 4a). Meadows with high to intermediate burial rates typically occurred in settings characterized by high autochthonous OC production at shallow depths (Fig. 4c–e), where moderately elevated MAR could promote rapid burial, limiting OC exposure to oxidation and enhancing long-term preservation. Log–log regressions of OC burial against sediment OC content within groups of sites binned by MAR revealed steeper slopes at higher MAR (Fig. 4b), suggesting that OC is more efficiently preserved in rapidly accumulating sediments[39]. In contrast, low burial rates were associated with environments characterized by limited fine-sediment deposition or low seagrass biomass, reflected in a larger offset between seagrass isotopic signals and that of sediment OC compared to high-burial settings (Fig 4c-e).

Sediments colonized by persistent seagrass species of the genera *Posidonia* and *Thalassia*, or by mixed meadows of persistent and opportunistic taxa (e.g., *Posidonia* with *Amphibolis*, or *Thalassia* with *Syringodium*), exhibited the highest sediment OC content (medians between 2.2% and 1.4%; $p < 0.05$; Fig. S2a), consistent with previous findings that persistent species support the largest sediment OC stocks globally[19]. While these genera exhibited high OC burial rates (Fig. S2c), burial was not strictly proportional to sedimentary OC content. Deviations from the mean OC burial-OC content relationship were structured by MAR (Fig 4b). At sites with low MAR, OC burial was lower than expected based on OC content alone, whereas at high-MAR sites burial exceeded values predicted by OC content.

This pattern was particularly evident in *Posidonia* colonized sediments, which, despite their highest OC content, did not achieve the highest OC burial due to characteristically low sedimentation rates (Fig. S2b, c and S3), reflecting their adaptation to thrive in environments with low sediment inputs, and consequently, low turbidity and high light availability [40]. In contrast, *Thalassia*-colonized sediments combined moderate to high OC content with elevated MAR, resulting in the highest OC burial efficiency (Fig S2c, S3). These observations were dominated by *Thalassia* beds in the tropical W. Atlantic, particularly in the lime-mud deposits of Florida Bay banks[41] and the Bahamas [42].

Sediments colonized by *Halophila* spp. in China[22], Australia[43], and the eastern Medtirerranean[44], showed some of the highest MAR, yet their low sediment OC content resulted in comparatively low burial rates. *Zostera* meadows displayed the widest range of both MAR and OC content among all genera (Fig S2), which may reflect the broad habitat envelope occupied by this genus relative to other seagrasses[45] and the possibility that *Zostera* and associated burial rates might not be well described using a single genus[46].

Coastal geomorphology is an important factor shaping global patterns of blue carbon stocks, particularly in mangroves[47], and new evidence suggest similar trends in seagrasses[19,36]. However, when assessing OC burial rates, global coastal geomorphology did not explain relevant differences in OC burial. Seagrass meadows in karstic systems exhibited the highest sediment OC burial rates, with these environments, also showing the highest MARs (Fig. S4). This pattern could be attributed to the limited representation of karst systems in other regions as cores in this category were predominantly associated with karstic settings in the Tropical W. Atlantic. Indeed, when the karst geomorphic category was excluded, or similarly, when cores from Florida Bay and the Bahamas were removed, coastal geomorphology had no significant effect on global OC burial rates. This pattern contrasts with the significantly higher seagrass OC stocks reported in arheic and small delta systems[19], which likely reflect longer-term accumulation rather than from high modern burial rates.

Organic carbon burial in vegetated and adjacent unvegetated sediments

About 18% of the cores in this study, including both vegetated and bare sediments, were collected in non-depositional environments, as indicated by negligible excess ^{210}Pb specific activities. Notably, the proportion of non-depositional records was twice as high in unvegetated sediments (30%) compared to vegetated ones (16%) (Fig. 5a), highlighting the role of seagrass in stabilizing sediments and OC stocks beneath their meadows[48]. However, whether depositional environments have a greater influence on OC burial than seagrass traits or the reverse, depends on the geomorphic and temporal context.

Seagrasses contribute to sediment stabilization by influencing local hydrodynamics, reducing flow speed, and ameliorating turbulence [49], facilitating the deposition of fine particles including both autochthonous and allochthonous OC[16,18,50]. Their network of roots and rhizomes further anchors sediments, mitigating resuspension and erosion [51] that could otherwise enhance OC remineralization. This stabilizing effect is likely most significant in settings where seagrass can reduce bottom shear stress below the critical threshold for sediment resuspension[52]. In higher-energy environments, or those already depositional, hydrodynamic conditions may independently govern sediment and OC burial regardless of vegetation cover[53].

Globally, sediments in non-depositional environments, although colonized by seagrasses, had significantly higher sediment dry bulk densities (DBD: 1.3–1.5 g cm^{-3}) and lower OC content (0.30–0.45%) than those in areas with measurable net deposition (DBD: ~0.85–1 g cm^{-3}; OC: 1.3%), irrespective of seagrass cover (Fig. 5b). Despite the differences in sediment characteristics between depositional and non-depositional habitats, the isotopic offset between seagrass and sediment OC ($\Delta^{13}C_{seagrass\text{-}sediment}$), which indicates the contribution of autochthonous and allochthonous sources, was similar across both habitats (Fig. 5b), pointing to a shared origin of sedimentary OC in depositional and non-depositional areas, even if their capacity to accumulate and retain OC differs.

At the local scale where seagrass meadows and adjacent bare sediments were sampled concurrently, vegetated sediments exhibited measurable sediment accumulation rates in similar proportions to unvegetated ones, with no significant differences in OC content, burial rates or in the relative contributions of allochthonous and autochthonous sources (Fig. 5c). Although limited in scope, these findings suggest that, within depositional environments, seagrass cover may not be a strong determinant of *in situ* OC burial. However, the presence of seagrasses contributes to the OC pool in adjacent unvegetated sediments, as evidenced by their similar isotopic signatures of seagrass-derived OC[18] (Fig. 5c), showing that seagrass OC can be redistributed across the depositional seascape, while long-term seagrass loss may reduce the overall OC pool. This is consistent with findings from the Bahamas[42] or Bermuda [54]. In the latter, sediment OC stocks were unrelated to seagrass abundance, yet seagrass loss led to a regional decline in seagrass-derived OC across both vegetated and unvegetated areas[54].

Local sedimentation dynamics, together with seagrass presence and OC export beyond meadow boundaries [18,55], can therefore result in similar OC burial rates between seagrass and adjacent bare sediments[41]. Unvegetated areas with measurable OC burial over the last century could thus represent both ongoing accumulation of seagrass OC or the legacy of past seagrass presence[56,57]. Together, these findings support a proposed hierarchy of controls on OC burial, with proximity to seagrass-OC sources as primary, hydrodynamic exposure as secondary, and vegetation traits as tertiary.

Global magnitude of organic carbon burial in seagrass sediments

A first order global OC burial rate in seagrass sediments over contemporary (~100-year) timescales can be estimated by scaling the area-weighted average from this analysis to the global seagrass extent. However, estimates of global seagrass distribution vary widely, ranging from 160,387 to 600,000 km^2 [45,58], while model-based projections suggest the potential extent could be an order of magnitude higher[59]. McKenzie et al.[58] revised the mapped seagrass area to 160,387–266,562 km^2, with high confidence in the lower bound and low confidence in the upper bound. We reviewed country-level seagrass coverage and integrated newly published distribution data since McKenzie et al.[58] (Table S4). Based on these updates, we conservatively estimated the current mapped seagrass area to range between 247,800 and 366,400 km^2. This updated estimate narrows the range of uncertainty while acknowledging gaps in coverage, particularly in parts of South and East Asia, which in turn support extensive meadows, the Arabian Peninsula, the Horn of Africa, North and West Africa, and the Baltic region. Except for the Arabian Peninsula, these gaps also extend to sediment OC burial rates.

Using these upper and lower area values and the area-weighted average OC burial rate, we estimate that global OC burial in seagrass sediments could range between 6 to 11 Tg OC yr^{-1} (assuming minimal extent) and 8–16 Tg OC yr^{-1} (assuming maximal extent). Previous global assessments used a broader seagrass area range of 300,000–600,000 km^2 [1,2,18], which would yield a mean global OC burial rate of 10–20 Tg OC yr^{-1}, still significantly lower than earlier estimates of 48–112 Tg OC yr^{-1} [2,18]. When sites with negligible sediment accumulation over the past century were included, the mean global estimate declined further to 5–7 Tg C yr^{-1} for the small area range, and to 6–11 Tg C yr^{-1} for the larger range.

Earlier estimates of global OC burial rates[2] incorporated net community production (NCP) estimates, a measure of ecosystem-level carbon balance, under the assumption that excess production was largely buried, or adjusted net primary production (NPP) values using the average fraction of OC storage derived from available budgets[14]. Seagrass ecosystems tend to be net autotrophic generating a large OC surplus, with an estimated NCP of about 119 ± 50 g OC m^{-2} yr^{-1} (36–71 Tg OC yr^{-1} globally)[17], which is not used by heterotrophs within the meadow but is either buried locally or exported away[14]. Assuming that roughly half of the global OC burial rate is supported by autochthonous production[18], our results suggest that 14 ± 7% of the NCP is retained and buried in situ within meadows, while the remaining ~85% may be redistributed to adjacent shorelines, unvegetated sediments, and offshore areas[55,60–62], where it could ultimately be buried, remain dissolved, or return to the atmosphere as CO_2.

Duarte and Krause-Jensen[55] synthesized existing estimates of earlier OC burial rates and seagrass ecosystem OC balance to show that about 24 Tg yr^{-1} of seagrass exported OC was sequestered beyond the meadow in unvegetated sediments or the deep sea. This exported fraction represents a substantial portion of the unburied NCP in situ and suggests that about 60% of the total seagrass NCP could ultimately be buried both within and beyond meadows. While earlier indirect carbon mass balance assessments by Duarte and Cebrián[14] already suggested a modest burial and significant export of seagrass production, our analysis reinforces and refines this view with updated burial estimates, further supporting the contention that the magnitude of exported and redistributed seagrass NCP is larger than previously recognized.

Seagrasses are the most widespread among blue carbon ecosystems[54,61,62], thus its contribution to the total OC burial in vegetated coastal sediments is large and accounts for ~30%, globally (Table S5). The total marine OC burial was revised by Duarte et al.[1] by adding the yet unaccounted contribution of seagrasses, mangroves, and tidal marshes to the long-standing estimates of marine OC burial[63,64]. We revisited this estimate with updated values of OC burial rates in seagrasses, mangroves[4] and tidal marshes[3] , as well as revised global burial rates for the continental margin[65] and deep sea[66]. Based on this approximation, we estimate that the OC burial within vegetated coastal sediments, including mangroves, tidal marshes, and seagrasses, accounts for ~8 to 13% of the total oceanic OC burial, with burial in seagrass sediments representing between 3-4% using the mapped seagrass area, or 3-6% assuming a $300\text{-}600 \cdot 10^3$ km^2 area range; a sizeable but more modest contribution than that assessed based on limited data a decade ago.

Uncertainties and needs for further research

Global patterns of seagrass OC burial rates across bioregions and coastal geomorphic settings remain poorly resolved, in part due to high variability in burial and sedimentation rates, and sediment OC content, which showed coefficients of variation of 75–100% across the dataset (Fig. S5). Most data on sedimentary OC burial rates come from Australia, USA and Spain (Fig. S6). When normalized for seagrass area, among countries with over 100 km^2 of seagrass, Canada, China, and Saudi Arabia showed the highest data density (Fig. S6c). Nevertheless, OC burial rates exhibited substantial spatial variability across countries even within well-sampled regions and at the site level, where variability remained high, at around 50% of the mean, highlighting the limited representativeness of existing datasets. This level of heterogeneity suggests the need for denser sampling and stratified designs that capture depth and regional gradients within countries to improve the accuracy of regional and national estimates. Countries with more than 10 observations had OC burial rate distributions that fell within our updated weighted average (~33 ± 10 g OC m^{-2} yr^{-1}) for Tier 1 within IPCC (Fig. 6a), including sites with negligible burial (Fig. 6b). In contrast, countries with fewer than 10 observations showed significantly lower rates, but the small sample size prevents determining whether these reflect true regional Tier 2 values or sampling limitations. Application of these data for Tier 2 carbon accounting (e.g., in the context of seagrass restoration under the IPCC Wetlands Supplement) remains premature without further representative sampling.

Conclusions

Our results support a downward revision of OC burial rates in seagrass sediments to 33 ± 10 g OC m^{-2} yr^{-1} (or 6–16 Tg OC yr^{-1} globally), providing a more accurate representation of the OC burial capacity that occurs within the sediments underlying seagrass meadows. While this fraction of the seagrass OC pool accounts for 3 to 6% of the total marine OC burial in the ocean, the significance of seagrass ecosystems as OC sinks may be larger. We estimate that seagrass meadows export a quantity of OC roughly three times greater than what is buried locally within the meadow, highlighting the importance of evaluating the fate and constraining the magnitude of this exported fraction to refine global estimates of seagrass contributions to oceanic OC storage. Moreover, the total seagrass area may be larger than that quantified here, and extensive historical losses of seagrass meadows[67] suggest that past contributions to global OC burial could have been larger, pointing to a potential for climate change mitigation by avoiding further losses and restoring seagrass ecosystems to their past extent[67,68].

Local sedimentary conditions play a prominent role in OC burial in seagrass meadows, and the relationship between sediment mass accumulation rates and sedimentary OC content can help delineate burial patterns, identifying sites with enhanced OC burial potential. Overall, the presence of seagrass serves as a reliable indicator of depositional environments conducive to OC burial, as non-depositional conditions were more frequently observed in unvegetated sediments. Within depositional settings both vegetated and bare sediments could exhibit similar OC burial dynamics under favorable sedimentary conditions. However, seagrasses remain critical for sustaining the regional OC pool over time, as their loss reduces the redistribution of seagrass-derived carbon across the landscape, ultimately diminishing long-term sequestration.

Methods

Data extraction and processing

We collated data from the available literature on direct estimates of OC burial rates over the past century based on ^{210}Pb, in seagrass-vegetated and adjacent unvegetated sediments, drawing from published articles, personal unpublished datasets, and gray literature. We compiled quantitative parameters such as ^{210}Pb specific activity profiles, derived mass accumulation rates (MAR; g cm^{-2} yr^{-1}) or sedimentation rates (SAR; cm yr^{-1}), sedimentary organic matter (OM) and OC content (%), and sediment dry bulk density (DBD). An effort was made to record core mass depths (g cm^{-2}) apart from DBD and OC content at each segmented interval; otherwise, mean values were recorded for these variables. Where available, ancillary data were recorded, including site and core coordinates, seagrass species and genera, $d^{13}C$ signatures of seagrass and sediments, sediment grain size, and ^{210}Pb profile quality (classified as ideal, surface mixing, downcore mixing, or negligible excess ^{210}Pb following Arias-Ortiz et al.[30]). Core locations were categorized into coastal geomorphological types based on their latitude and longitude using the global classification scheme by Dürr et al.[34]. Each core was assigned to one of the following categories: small deltas (I), tidal systems (II), lagoons (III), fjords (IV), karstic coasts (VI), or arheic (dry) coasts (VII).

Glaciated coasts (0) and large rivers (V) were not represented in the dataset. Compaction of sediments may occur during coring and affect depth and DBD measurements and, consequently, OC accumulation rates when these are derived from OC density and vertical accretion (SAR). Where reported, we recorded evidence of compaction and noted whether sedimentation rates had been corrected accordingly.

When available, we used published OC burial rates directly. For studies where rates were reported for specific depth intervals, dates predating the excess ^{210}Pb horizon, or where only radiometric dates (but not accumulation rates) were provided, we recalculated them to represent the mean OC burial rate over the past century using the full-dated profile. To minimize the effects of possible core compaction, we recalculated OC burial rates as the product of the fraction of OC and the MAR (Eq. 1), as opposed to the product of OC density (g OC cm^{-3}) and the sediment accretion rate (cm yr^{-1}), which are affected by compaction artifacts. This approach was applied in studies where either original rates required recalculation (cases above) or the reported SAR-based estimates did not account for compaction.

$OC\ burial\ rate = C_t \cdot MAR_t$ (Eq. 1)

where C_t is the fraction of OC accumulated down to the excess ^{210}Pb horizon, and MAR*t* is the mean mass accumulation rate of that period.

Restored seagrass sites were not included in our analysis (e.g. restored meadows in refs.[57,69–71]) as their OC burial rates might be affected by the time since planting, which varies between sites and does not align with the temporal resolution over which OC burial rates were assessed in the rest of the dataset's cores.

The full dataset comprised 326 cores from 64 studies, including 45 published articles, 3 theses and 3 technical reports, and 13 unpublished datasets. In contrast to recent reassessments of OC burial rates in tidal marshes[3,27] and mangroves[4,28], we also considered sites where intense downcore mixing or negligible excess ^{210}Pb accumulation precluded the determination of contemporary OC burial rates. Sites exhibiting intense downcore mixing (> ½ of the datable part of the profile), but where OC burial rates were reported (25% of cores), were flagged. Global areal rates were then computed with and without these values to assess the potential overestimation of OC burial rates derived from ^{210}Pb due to the well-known effects of sediment mixing in biasing sedimentation rates high. Sites exhibiting intense mixing where OC burial rates could not be resolved were retained in the dataset but rates were recorded as NA. At sites where ^{210}Pb specific activities reflected only the background (supported levels of ^{210}Pb) suggesting no net sediment accumulation (18% of cores), we assumed that the net OC burial over the last century was negligible and not significantly different from zero. By factoring these sites into our analysis, we estimated a lower bound for areal OC burial in seagrass sediments that is less biased toward depositional environments.

In addition to vegetated seagrass cores, we also included unvegetated sediments sampled alongside seagrass meadows in the reviewed studies. These were treated separately but classified and analyzed using the same criteria and methods. At sites where paired data were available (17 sites, of which 8 exhibited net positive sediment accumulation over the past century), we directly compared OC content, $D^{13}C_{seagrass - sediment}$ offsets, sedimentation and OC burial rates between vegetated and unvegetated sediments.

Ten studies (56 cores) estimated OC from organic matter (OM) content using local site-specific regressions, and four studies used the general equation developed by Fourqurean et al.[20]. In nine cores sampled from the banks of Florida Bay[41], OC content was neither measured nor estimated, although OM content was provided. A site-specific equation published for Florida Bay[72] was applied to these sites.

Global OC burial rate estimation and area scaling

Global OC burial rates per unit area were estimated using two approaches. First, we calculated the geometric mean of OC burial rates for vegetated sites with net positive sediment accumulation, and the median across all vegetated sites, including those with negligible accumulation, as these metrics best represent the central tendency. Second, an area-weighted average was calculated using the relative contribution of each seagrass bioregion to the total seagrass extent. For the latter calculation, estimates of OC burial rates were grouped by seagrass bioregions described in Short et al.[33] (i.e., Temperate North Atlantic, Temperate North Pacific, Mediterranean, Tropical Atlantic, Tropical Indo-Pacific, and Temperate Southern Oceans). In our analysis, the Tropical Atlantic is represented primarily by the Tropical Western Atlantic, with only two cores available for the eastern side of this bioregion. Similarly, estimates for the Temperate Southern Oceans are largely derived from Australian sites, reflecting limited data availability from other parts of the Southern Hemisphere. Then, the geometric mean OC burial rates per unit area were estimated for each bioregion, and globally as an area-weighted average based on the relative contribution of each bioregion to the total global seagrass extent. Cores with negligible sediment accumulation were excluded from the weighted average estimate because such observations are not reported as consistently as those with net positive accumulation, often due to perceived ^{210}Pb dating failures. Including these sites would disproportionately penalize bioregions where authors more frequently report lack of accumulation based on ^{210}Pb dating. Moreover, previous global syntheses have not incorporated these sites, ensuring consistency for comparative purposes. Nevertheless, for global summary statistics across the full dataset, we consider it important to include sites with little-to-no sedimentation, as negligible excess ^{210}Pb is itself a meaningful result that may reflect sediment dynamics rather than merely methodological limitations and its reporting should be encouraged.

Over a 3-fold bracket exists in the global area reported for seagrass meadows (160,387–600,000 km^2)[45,58,73]. The lower estimate is based on the moderate- to high-confidence mapped area and is widely acknowledged as an underestimate[45,58]. For this study, we revised the mapped seagrass area in McKenzie et al.[58] by incorporating more recent estimates published since (Table S4). For countries where updated seagrass area estimates were available, we subtracted the previously assigned areas in

McKenzie et al.[58] and added the new figures. These updates resulted in a revised mapped seagrass area range of 247,800–366,400 km^2. A substantial portion of the increase from the earlier range reported by McKenzie et al.[58] (160,387–266,562 km^2) is attributable to the recent mapping of extensive (> 57,000 km^2) seagrass meadows in The Bahamas[74], and in Insular Caribbean[75] among other additions. Previous assessments of OC burial rates in seagrass meadows commonly assumed a global extent of 300,000–600,000 km^2 [1,2,18]. To enable comparison with these studies, we also upscaled OC burial rates using this range, alongside our updated mapped seagrass extent.

Statistical analysis

We analyzed the distribution, skewness, and kurtosis of the data. A Shapiro-Wilk test provided an indication of normality for the untransformed and transformed versions of the dataset. These results were used to assess which parameter (the arithmetic mean, median, or geometric mean) was best suited to represent the central tendency of our data. The arithmetic mean was used if data followed a normal distribution, the geometric mean if a skewed distribution was made symmetrical by a log transformation, and the median if either the untransformed or log-transformed dataset did not follow a normal distribution. Non-parametric tests were generally used for statistical comparisons, as most parameters did not follow a normal distribution. Paired two-group comparisons (e.g., between vegetated and adjacent unvegetated sites) were assessed using the paired-sample Wilcoxon signed-rank test, while differences across multiple groups (e.g., among bioregions, coastal typologies, or seagrass genera) were tested using the Kruskal–Wallis test followed by Dunn's post hoc test with Bonferroni correction. All statistical analyses were done at a level of significance of $\alpha < 0.05$. To examine controls on OC burial, we analyzed relationships between sediment OC content, MAR, and OC burial rates across vegetated sites. Burial rates were grouped into quantile intervals (low: 0–25%, intermediate: 25–75%, high: 75–100%) to identify thresholds associated with contrasting depositional settings. Log–log regressions were used to assess how OC burial scales with OC content across MAR quantiles, and slopes were compared to interpret sediment OC preservation under varying sedimentation rates.

Declarations

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Data availability

Sediment core datasets are available through the Smithsonian's Figshare data repository: doi pending.

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Figures

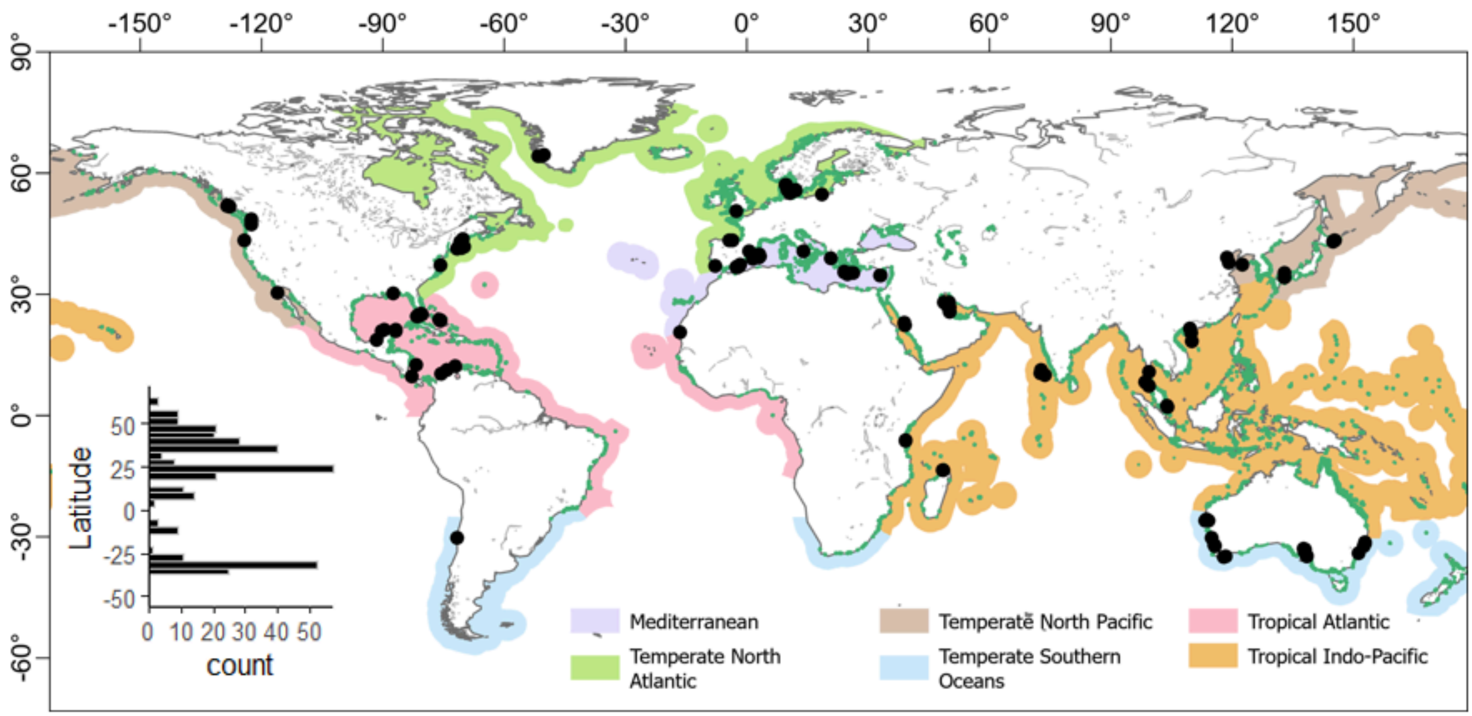

Figure 1

Location of sites included in the synthesis superimposed on global seagrass bioregions. Black dots represent sediment cores. Dark green coastline indicates the global seagrass distribution[37]. The inset bar graph shows core counts by 5° latitudinal bands.

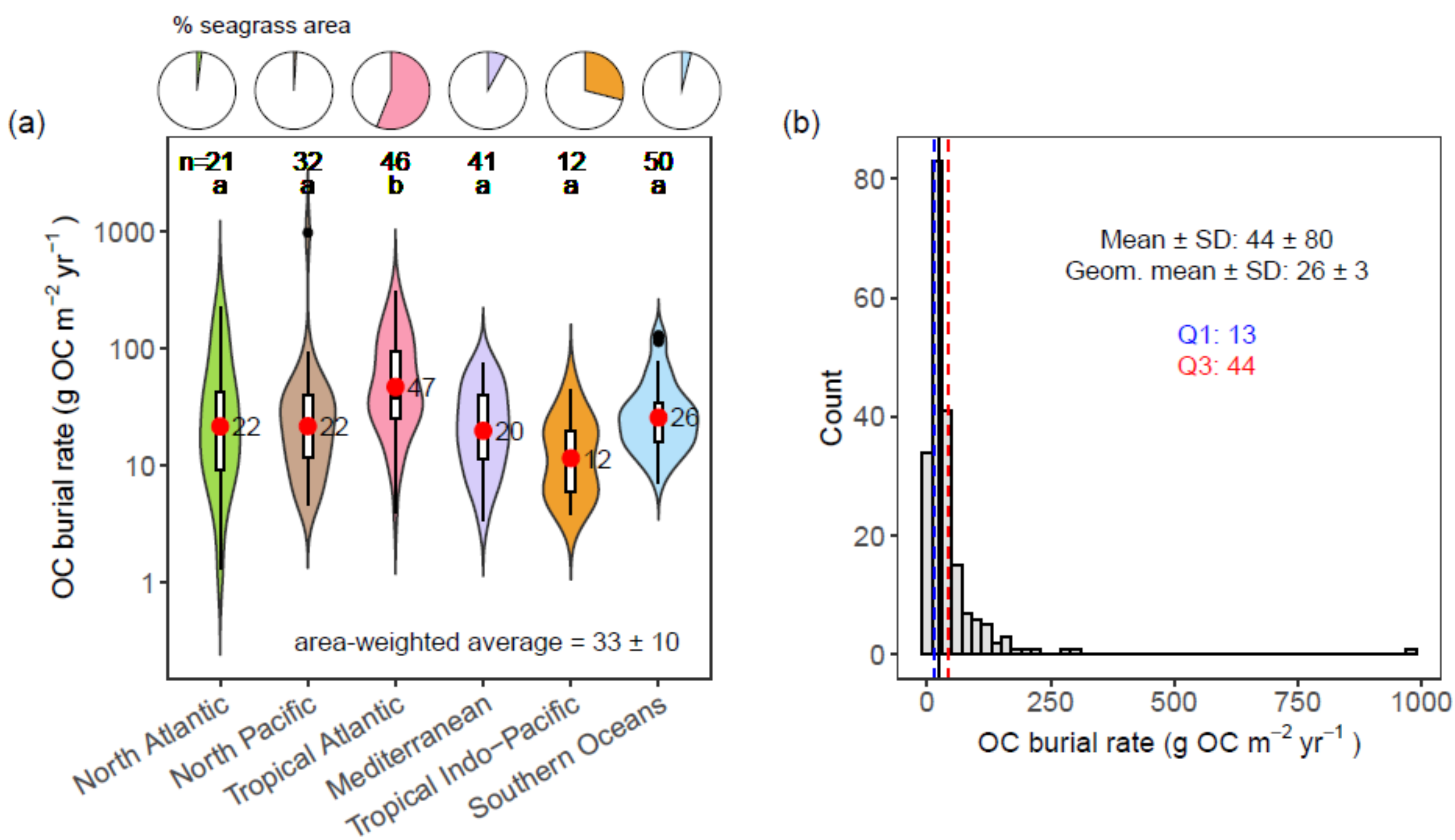

Figure 2

Organic carbon (OC) burial rates in seagrass sediments with net positive accumulation. (a) OC burial rates by bioregion with red dots indicating the geometric mean. (b) Binned frequency of all OC burial rates in vegetated sediments. Pie charts in panel (a) show the percentage contribution of each bioregion to the total seagrass area (Table S3). The area-weighted average was calculated using these proportions as weights. In panel (a), different lower-case letters represent significant differences between bioregions as a result of a Kruskal–Wallis Dunn's multiple comparison test with $p < 0.05$ adjusted using the Bonferroni procedure ($p < \alpha/2$). Numbers above letters denote the number of cores per bioregion. Q1 and Q3 in panel (b) represent the first and third quartiles, indicating burial rates for the bottom 25% and 75% of the dataset, respectively.

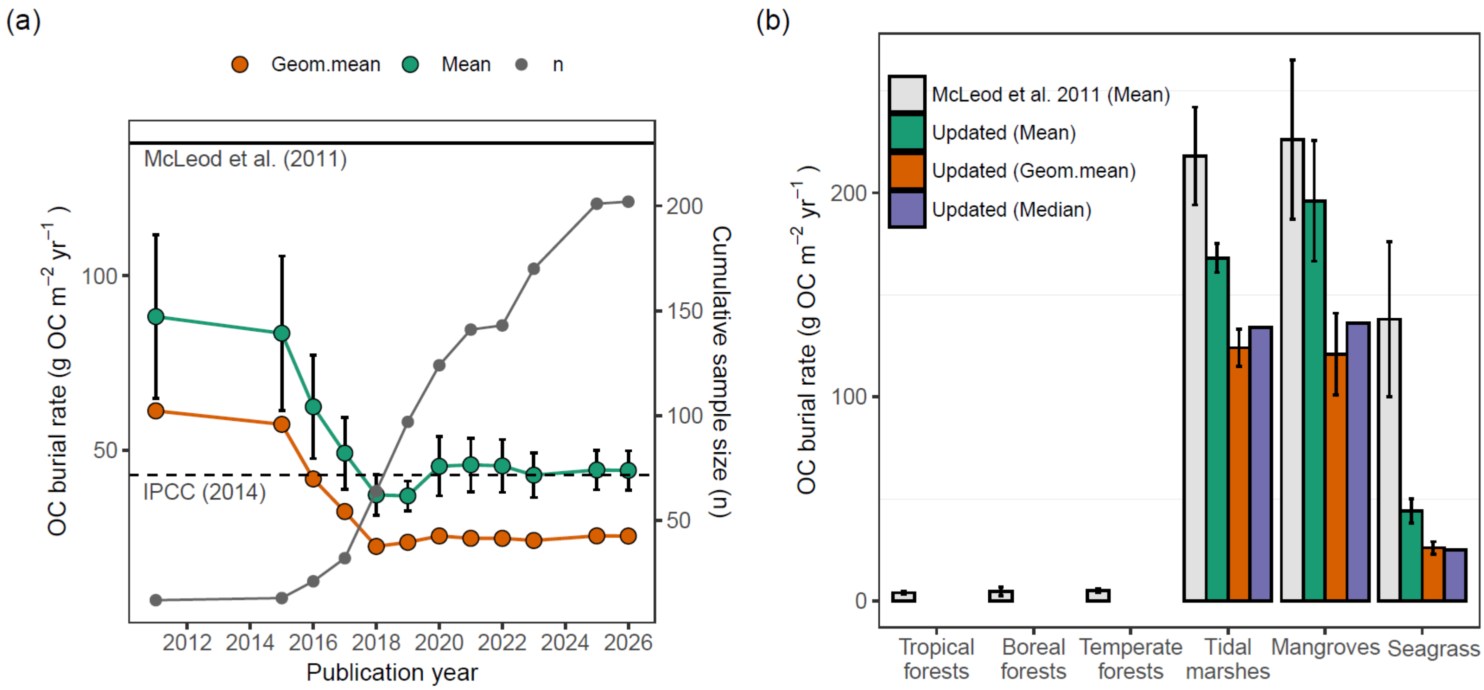

Figure 3

Summary statistics of contemporary organic carbon (OC) burial rates in seagrass sediments. (a) Cumulative summary statistics of OC burial rates by publication year, calculated from 2010 onward, for seagrass cores exhibiting net positive accumulation (b) comparison with OC burial rates in soils of terrestrial forests, sediments of other coastal vegetated ecosystems and previously compiled estimates. In panel (a) median estimates are not shown because they overlap with geometric mean values. Equivalent statistics for both depositional and non-depositional seagrass sediments are presented in Fig. S1. Horizontal lines indicate the IPCC Tier 1 estimate (dashed) and the mean OC burial rate reported by McLeod et al. (2011) (solid). In panel (b), results from previous and updated assessments are shown, with revised estimates from Breithaupt and Steinmuller (2022) for mangroves, Wang et al. (2021) for tidal marshes, and this study for seagrasses. Error bars represent the standard error of the mean, and the distance from the central tendency to the bounds of the 95% confidence interval for the updated geometric mean estimates.

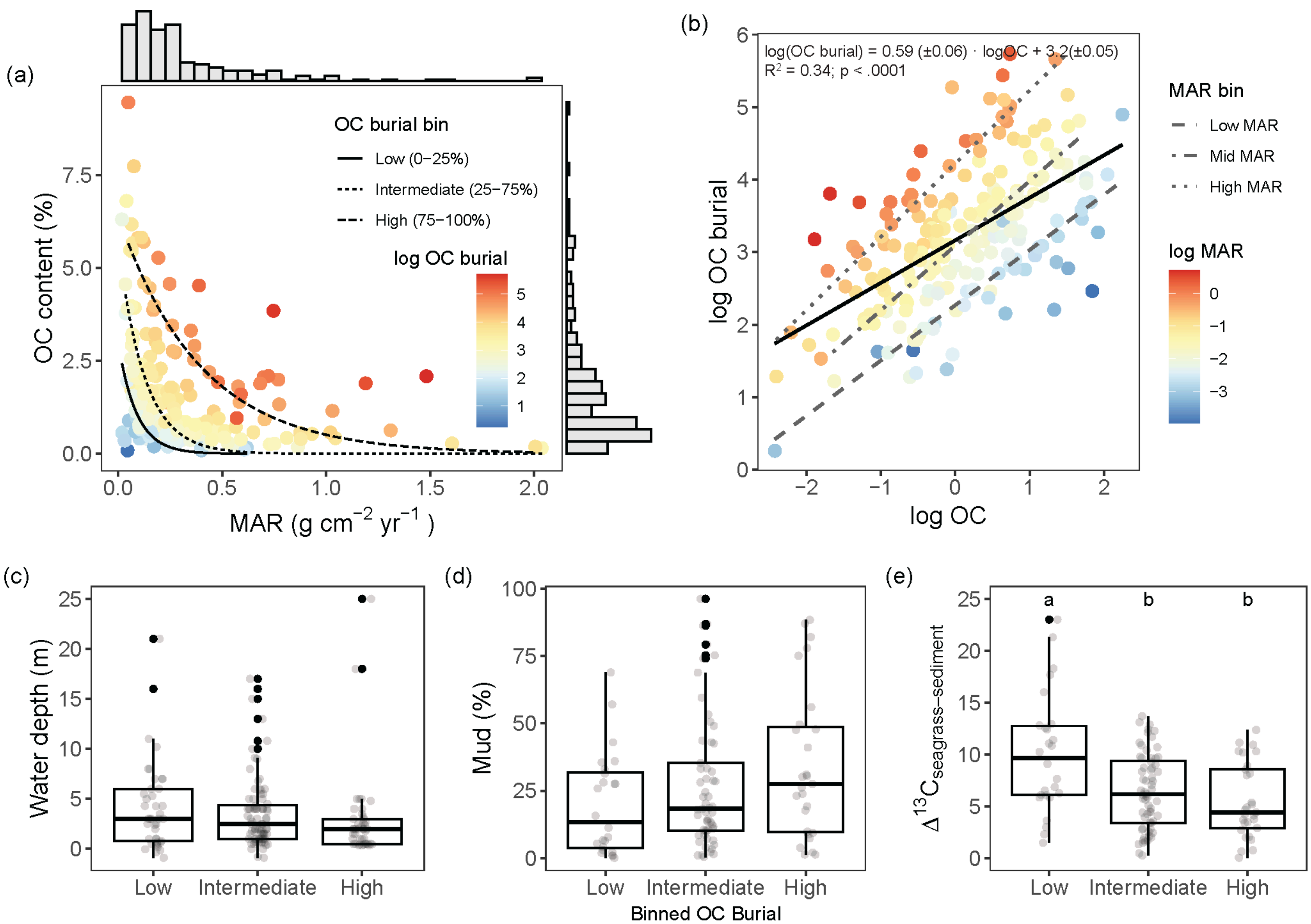

Figure 4

Environmental and geochemical factors associated with organic carbon (OC) burial rate in seagrass sediments. (a) Relationship between sediment OC content and mean mass accumulation rate (MAR) at vegetated seagrass cores. The color gradient represents OC burial rates on a log scale, with exponential fits showing varying slopes of the relationship for binned burial rates, grouped by percentiles (0–25%, 25–75%, 75–100%). Regression statistics are summarized in Table S3. Histograms on the top and right show marginal distributions of MAR and OC, respectively. (b) Log-log relationship of OC burial and OC content in vegetated sediments. The color gradient represents MAR on a log scale, with dashed lines showing linear fits for each MAR quantile (0–25%, 25–75%, 75–100%) and a solid line representing the overall mean linear fit across all points (equation shown; quantile regression stats in Table S3). **(c–e)** Boxplots showing environmental and geochemical characteristics across OC burial bins (Low, Intermediate, High): **(c)** water depth, **(d)** mud content, and **(e)** isotopic difference between seagrass and sedimentary OC. Boxes show the interquartile range, horizontal lines indicate the median, and whiskers extend to 1.5 times the IQR, individual data points are overlaid and solid dark data points represent outliers. Mass accumulation rates larger than 3 g cm^{-2} yr^{-1} and OC burial larger than 500 g C m^{-2} yr^{-1}

(n=1) have been excluded as outliers. Significant differences in environmental or geochemical characteristics between OC burial bins in the boxplots are indicated with letters.

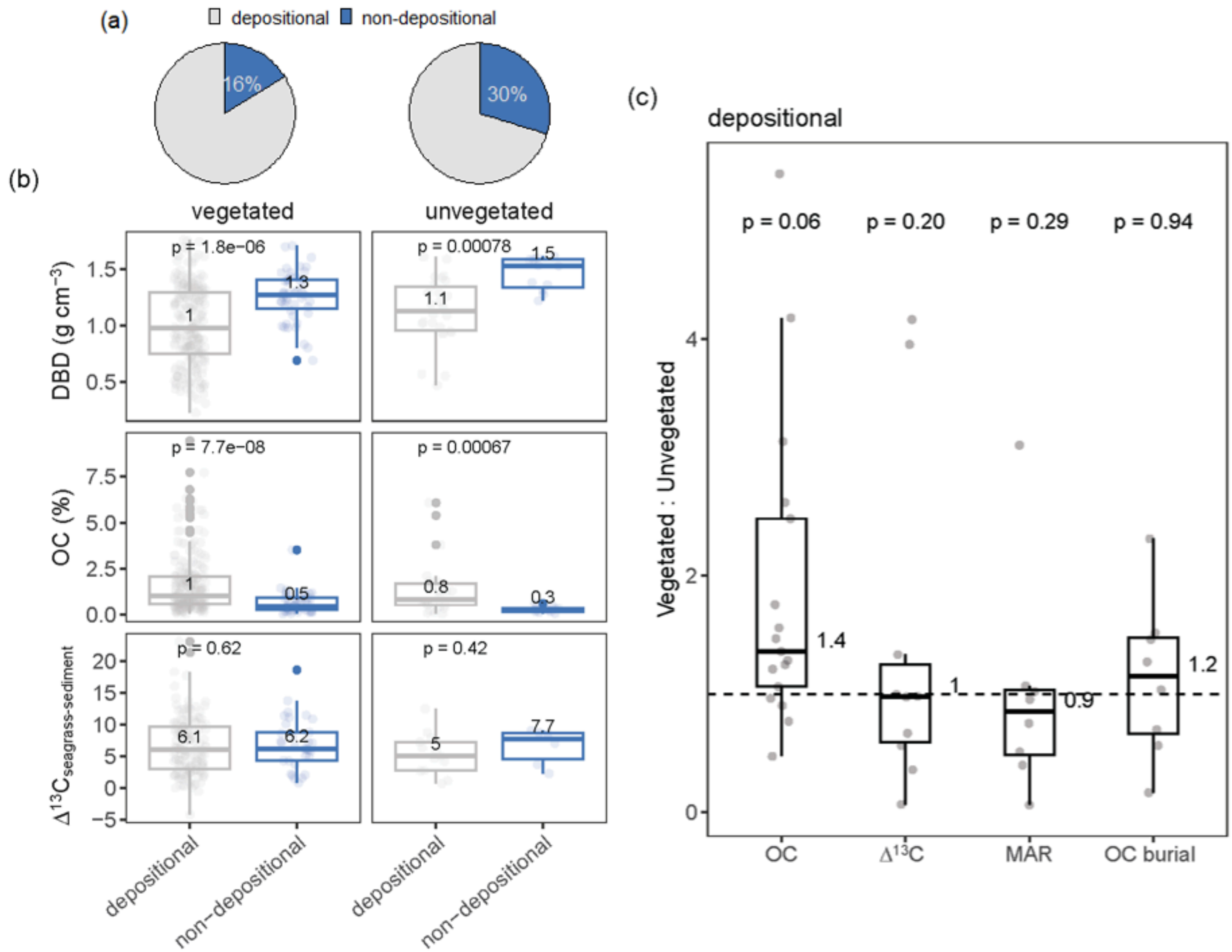

Figure 5

Comparison of depositional and non-depositional sites across vegetated and bare seagrass sediments. (a) Proportion of depositional and non-depositional records within vegetated and unvegetated sediments, where grey represents depositional and blue non-depositional records. (b) Comparison of sediment properties and contributions of autochthonous organic carbon (OC) between depositional and non-depositional sediments in vegetated and unvegetated habitats. (c) Boxplots of the ratios of DBD, OC content, $D^{13}C_{seagrass\text{-}sediment}$, mass accumulation (MAR), and OC burial rates between paired seagrass and unvegetated sediments, grouped by site. DBD stands for sediment dry bulk density, and $D^{13}C_{seagrass\text{-}sediment}$ reflects the difference between the $d^{13}C$ isotopic signal of seagrass plant material and that of the sediment. In panel (c), significant differences between paired samples were analyzed using a paired t-test (DBD, and $D^{13}C_{seagrass\text{-}sediment}$) and a Wilcoxon signed-rank test (OC content, MAR, OC burial) (p <

0.05). Paired comparisons in panel c included 17 sites for DBD and OC content; 10 for $D^{13}C_{seagrass-sediment}$, and 8 observations for MAR and OC burial.

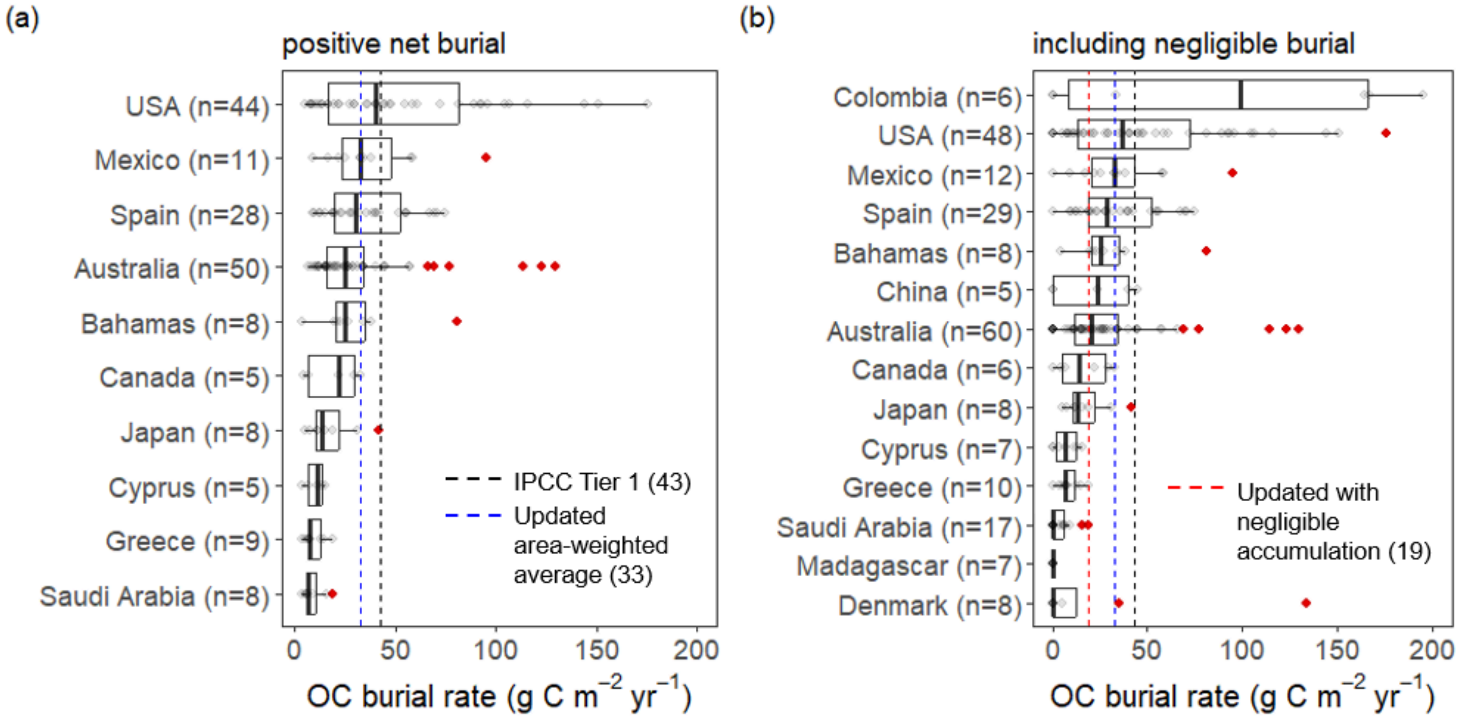

Figure 6

Boxplots of organic carbon (OC) burial rates across countries under (a) Vegetated cores showing net positive OC burial rates, (b) including cores with negligible burial rates. Points represent individual cores. Countries are ordered by their median OC burial. Only countries with five or more observations are shown. The dashed black and blue vertical lines indicate, respectively, the current IPCC Tier 1 value (43 g OC m^{-2} yr^{-1}) and updated OC burial rate based on the area-weighted average (33 g OC m-2 yr-1) and median when negligible accumulation was considered (19 g OC m^{-2} yr^{-1}) . Values above 200 g OC m^{-2} yr^{-1} are considered but not shown for visualization purposes. Red diamonds show outliers.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- AriasOrtizetalsuppotingv3.docx
- TableS4.xlsx